NHS: Capacity and Sentiment Analysis

Context

The National Health Service (NHS) is the UKs public healthcare provider and is both essential to the welfare of UK citizens and part of the fabric of the UK's identity. The NHS's key problem is that it requires scaling to match the growing, aging population of the UK - all with a limited budget. Notable for the time period, is that the NHS also faced the additional strain of the Covid-19 pandemic.

NHS scale can be achieved through either investment and growth to increase capacity, or by improving efficiencies. This analysis utilises the provided datasets to suggest actions in response to the questions posed by the NHS regarding adequate staffing, capacity and resource utilisation. As analysts, we also ask our own questions and challenge the usefulness of the data.

Analytical Approach

The first step is to import pandas and numpy alongside the datasets. We then perform Descriptive Analysis starting with a sense check() and perform descriptive statistics. There are no missing values and each dataset varies in scale with each appointment dataset containing varying levels of 'Count of Appointments'.

We identify 107 locations, with each month varying in the number of appointments, North West London being the largest and Greater Manchester the smallest. Key variables include: 'National_Cateogry', 'Service_Setting', 'Context_Type' and 'Appointment_Status', with there being respectively 18,4, 3 and 3 number of measures in each.

On deciding mapping and variable exclusion, we opt to only map 'National_Category' and provide a list for mapping, grouping the number of variables by theme from 18 to 11. In our other key variables, we do not map or exclude 'missing data' measures such as 'Unmapped' and 'Other' as these metrics are important in telling the story of the NHS's effectiveness in data collection.

Stand out values include: General Practice being the most popular 'Service_Setting' with 359,274 counts. As for 'Appointment_Status', there are a significant number of 'Unknown' appointments, 37,964 more than 'DNA'. As assumed, 'Face to Face' is still the largest 'Appointment_Mode' however, surprisingly, Telephone appointments are close at only 8% less.

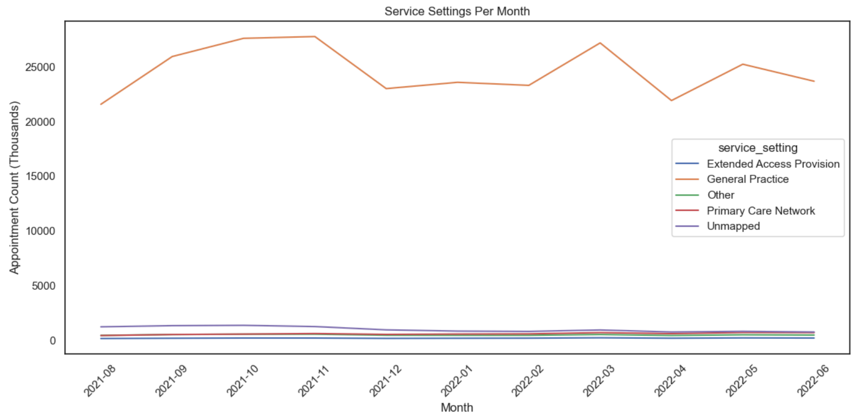
Importantly, from identifying each dataset's timeframe, we pinpoint a core timeframe where the three datasets have overlap in appointment times: from December 2021 to June 2022.

Later in the notebook, to aid visualisation and analysis we perform additional transformations. These include importing Seaborn, dividing appointment count by 1000 to make the Y axis of visualisations legible and we transform 'Appointment_Status' into a percentage to measure the percentage of appointments attended. Lastly, research found that on average a missed NHS appointment costs the NHS £30, thus we use this to monetise missed appointments.

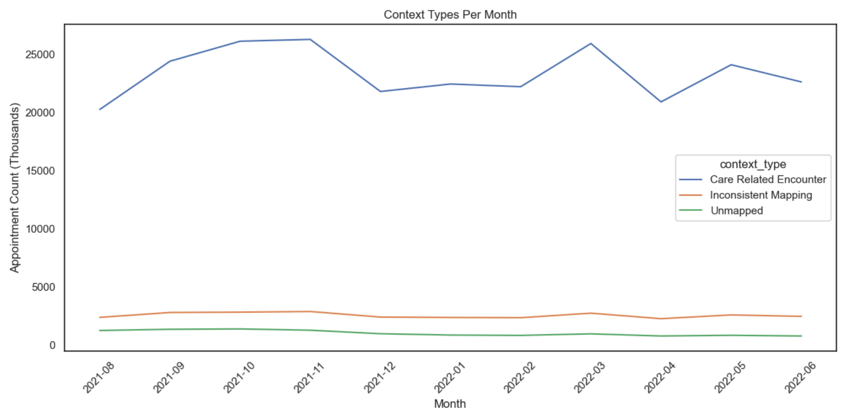
Finally, where appropriate, we convert date fields into strings to allow for visualisation.

Capacity Analysis

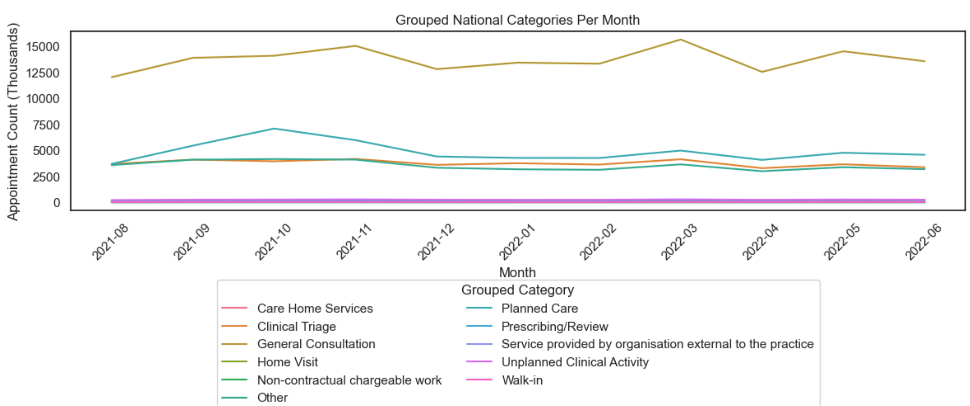
1a)



1b)

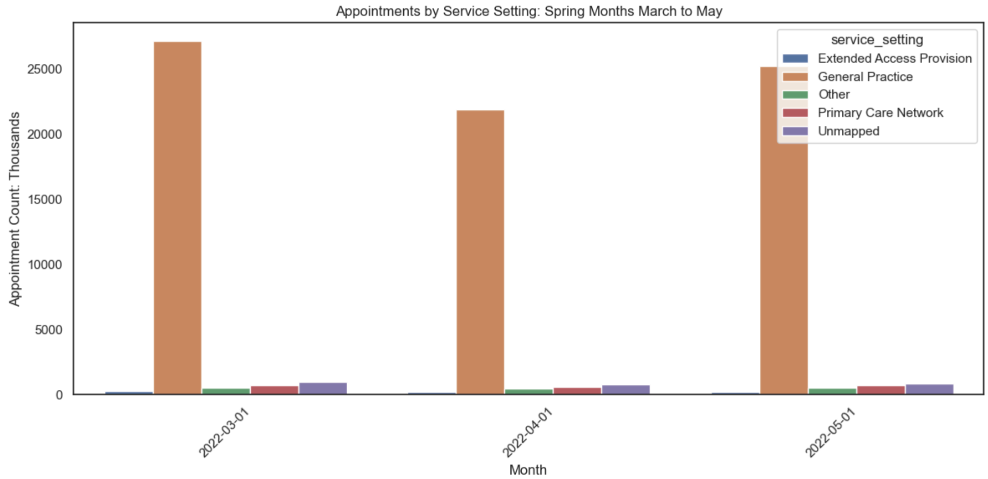


1c)



Onto plotting the data, the key measures follow the same seasonal trends where there is an increase or peak in appointments in October and November 2021, and another peak in March 2022. The categories with less appointments are less legible - we choose to not make further distinction within the plots to maintain the clarity of the dominating category such as ‘General Practice’. Importantly, for National Categories, we apply our grouping(1c) and across all plots where Appointment Count is the Y axis, appointment counts are read in thousands.

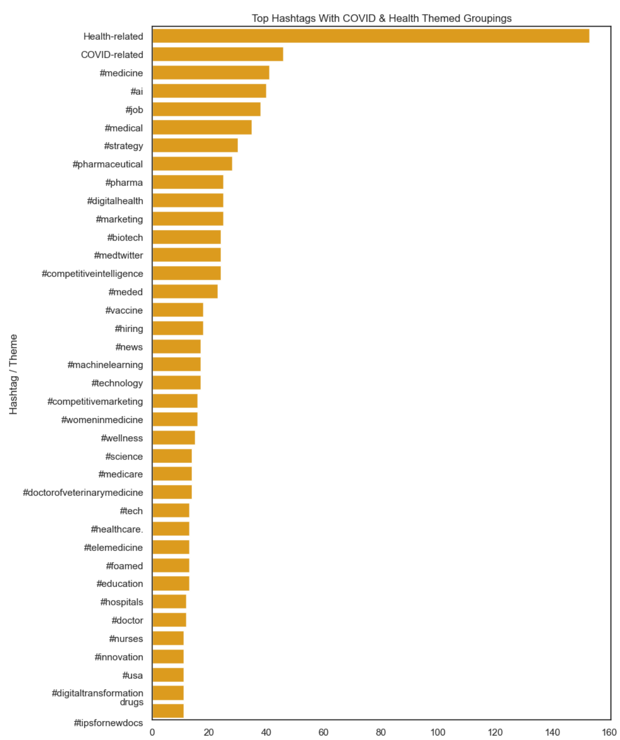
1d)



Bar charts are used to analyse seasonal trends of Service Setting, as it is not possible to line plot the Summer season for only August has data. For the Spring Months, we can see a slight decrease in the 'Unmapped' category (1d).

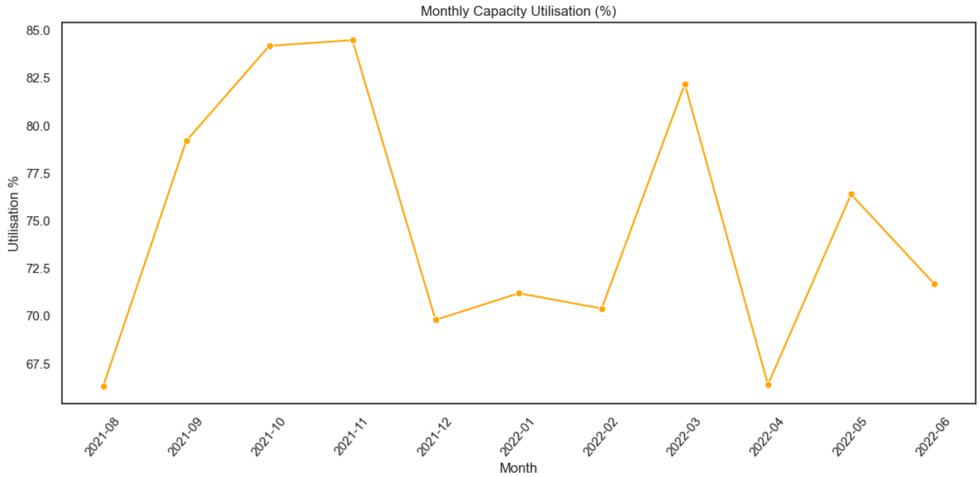
Sentiment Analysis

2a



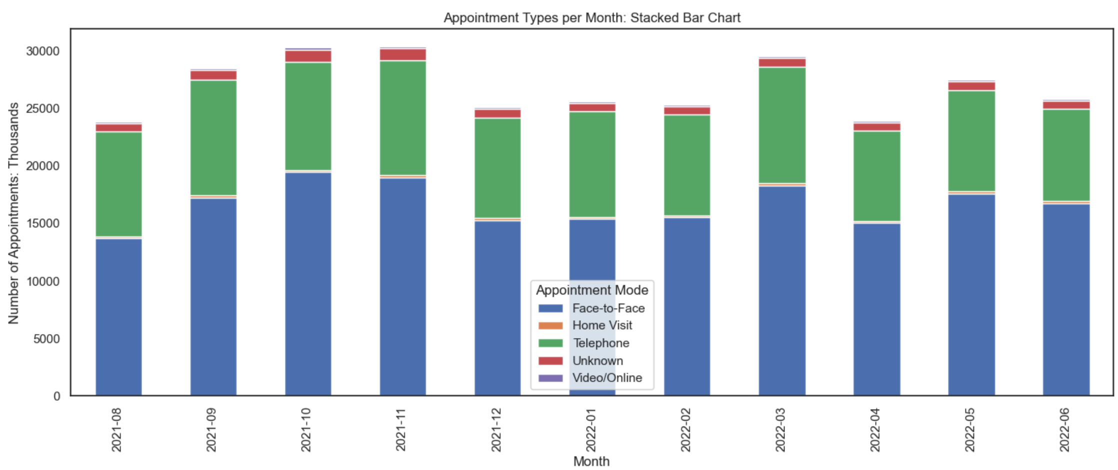
We present the Twitter analysis in bar chart form where we remove overrepresented results and map themes(2a). The most interesting hashtag grouping was the 'Covid' tag, the most frequent illness related hashtag, indicative of the challenge the NHS faced at the time of data collection. We also look at top retweets and favourited tweets, however find mainly adverts and entertainment content, less useful to analyse.

3a

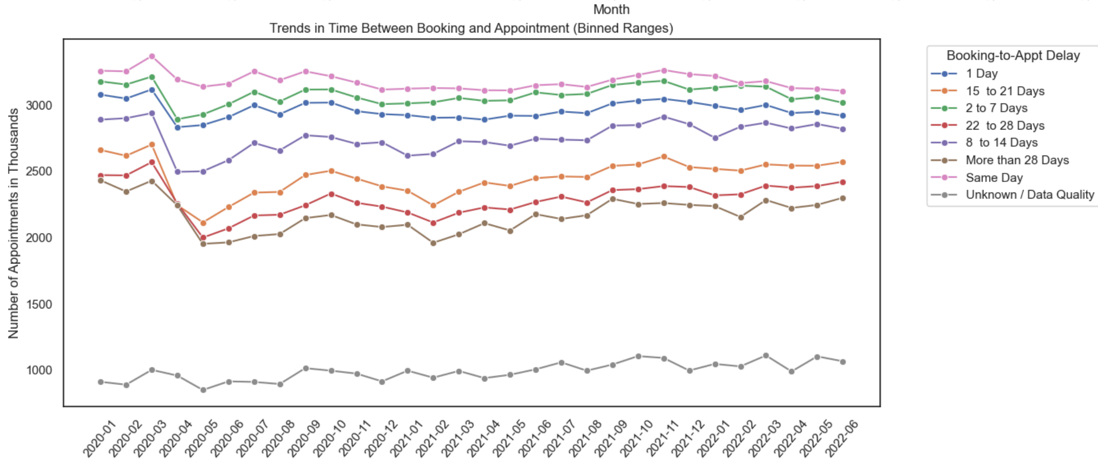


Line plots are used to present findings on staff level utilisations where it peaks at 85% in October and November (3a), meaning there was still capacity in the system in the busiest months. There is a 3.5% range in the highest and lowest Appointment Attendance. This is a small numeric difference but a large cost to the NHS when scaled across the thousands of appointments. With that, 'unknown' still accounts for around 5% of the data, negatively impacting data quality.

3b)

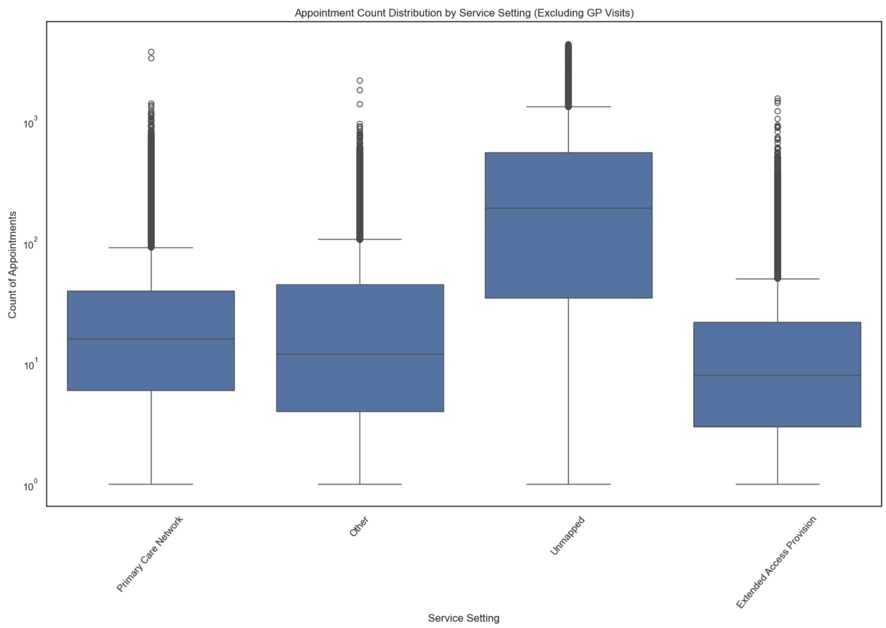


3c)



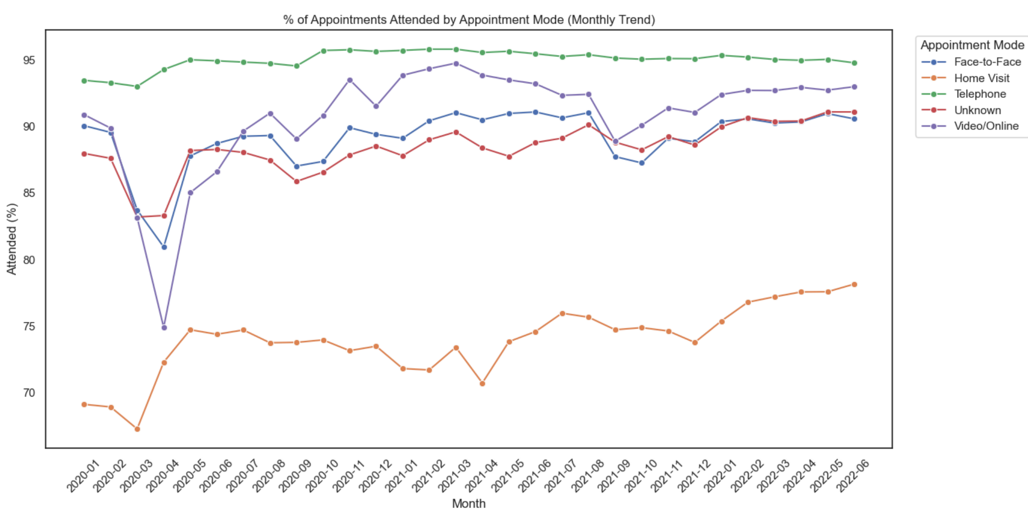
The appointment type split remains proportionate across the busiest months, as displayed by a stacked bar graph (3b). Time Between Booking and Appointment Rate remains consistent until March 2022 where the count of appointments with lower waiting times reduces,also increasing the count of higher waiting times (3c) - a sign that resource strain is increasing.

3d)



Finally, to compare the Service Settings we plotted a boxplot and adjusted to logarithmic scale to allow easier readability. General Practice has the widest range and highest level of outliers. Once General Practitioner is combined, unmapped has the largest range (3d).

3e)



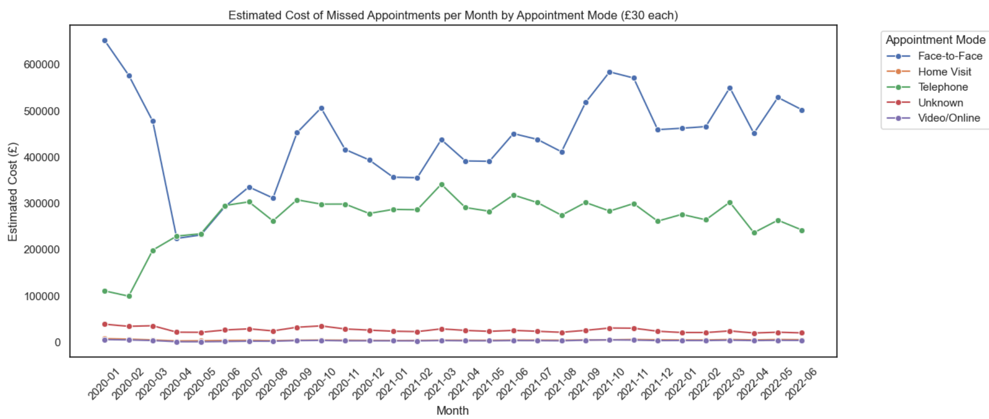
The Appointment Type with the highest Attended Rate was ‘Telephone’ (~95%) and the lowest was ‘Home Visit’ (~75%). We show the longer the period between the appointment being booked and the planned appointment day, the less likely individuals are to attend (3e), costing the NHS a significant amount.

Notable Patterns, Trends and Recomendations

Regarding notable patterns and trends, there is a consistent increase in appointments in the months of October, November which we hypothesise to be from the natural downturn in weather and March, due to a backlog of appointments from the Omicron Covid-19 variant. Staff utilisation is consistent, however for this analysis to be valuable, data must be evaluated over several years.

Hence the next step would be to mimic this analysis across several years of data to account for a period where the data is not influenced by a worldwide pandemic. We also recommend improving data collection to reduce the number of 'Unknown' data records.

4a)



Our core recommendation is to move more appointments to video / online to help save the NHS money in missed appointments (4a), with the saved money being reallocated to increased staffing, to help reduce the waiting time, thus further reducing missed appointments.